

Integrated Retail Analytics for Store Optimization and Demand Forecasting

Project Report

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Google Colab Notebook: [Google Collab Link](#)

GitHub Repository: [Github Link](#)

1. Executive Summary

This project analyzes weekly retail sales to support store performance review, sales forecasting, and business planning. Three datasets were combined: sales, store-week features, and store attributes.

The sales dataset contains 421,570 observations covering 45 stores and 81 department identifiers from 5 February 2010 to 26 October 2012.

The analysis includes data validation, exploratory visualization, feature engineering, anomaly detection, store segmentation, department association analysis, and forecasting at one-week and thirteen-week horizons.

For one-week-ahead forecasting, the selected gradient boosting model achieved a test MAE of 1,276.44, RMSE of 2,652.42, and WAPE of 8.06% on 38,009 observations shared with the baselines. Its MAE was approximately 16.3% lower than the last-week baseline. Adding lagged external features did not improve validation MAE under the tested settings.

For thirteen-week-ahead forecasting, the seasonal baseline achieved the lowest validation MAE and was selected. Although the direct boosting model achieved a lower validation RMSE, MAE remained the predefined selection criterion.

Store segmentation identified two groups with different average sizes, sales levels, and relative sales variability. These findings support differentiated planning reviews and controlled marketing pilots.

Customer identifiers and transaction baskets are unavailable. Therefore, the segmentation describes stores, and department associations represent sales-pattern hypotheses rather than evidence of products purchased together.

2. Business Problem and Objectives

Retail sales vary across stores, departments, and weeks. Understanding these differences can support sales planning, store performance review, and investigation of unusual activity.

This project addresses the following business questions:

- How do sales differ across stores, departments, and time?
- What sales patterns appear around holidays and across months?
- Which sales changes are unusual compared with historical behavior?
- Can stores be grouped using similar sales characteristics?
- Do strong-sales events in different departments occur together?
- How accurately can sales be forecast one and thirteen weeks ahead?
- Do historical economic, weather, and markdown variables improve forecasting beyond sales history and calendar information?

The objectives are to build a reproducible analytical workflow, compare forecasting approaches against simple baselines, evaluate store segments, and develop practical recommendations supported by the findings.

The forecasting target is weekly sales amount, not product quantity. Exact reorder quantities and safety stock require additional data, including unit demand, available inventory, supplier lead times, and service-level requirements.

Individual customer personalization cannot be evaluated using these datasets. Marketing recommendations therefore focus on store-segment pilots that require controlled testing before wider implementation.

3. Dataset Description

The project uses three CSV datasets. Together, they provide weekly department sales, store-week contextual information, and store attributes.

Sales Dataset — sales data-set.csv

This dataset contains 421,570 rows and 5 columns: Store, Dept, Date, Weekly_Sales, and IsHoliday.

Each row represents one store-department-week observation. Sales dates range from 5 February 2010 to 26 October 2012, covering 143 weeks, 45 stores, 81 department identifiers, and 3,331 distinct store-department combinations.

The forecasting target is Weekly_Sales, which represents a sales amount rather than the number of products sold.

Features Dataset — Features data set.csv

This dataset contains 8,190 rows and 12 columns. Each row represents one store-week observation.

It includes Store, Date, Temperature, Fuel_Price, Markdown1 through Markdown5, CPI, Unemployment, and IsHoliday. Its dates extend from 5 February 2010 to 26 July 2013.

The five markdown variables contain substantial missing information. Missing values are not assumed to mean that no promotion occurred. The exact markdown categories and measurement units require confirmation from source documentation.

Stores Dataset — stores data-set.csv

This dataset contains 45 rows and 3 columns: Store, Type, and Size. Each row represents one store.

There are 22 Type A stores, 17 Type B stores, and 6 Type C stores. The detailed category definitions and store-size units are not inferred without supporting documentation.

Dataset Relationships

Sales and features are linked using Store and Date. Store attributes are linked using Store.

Multiple department sales records can share one store-week feature record. Therefore, the joins are validated as many-to-one relationships to prevent accidental duplication.

Data Limitations

The datasets do not contain customer identifiers, transaction baskets, product names, explicit store locations, inventory levels, supplier lead times, or unit demand.

These limitations restrict customer-level personalization, true market basket analysis, geographic interpretation, and precise inventory optimization.

4. Data Validation and Preprocessing

Date Conversion and Record Validation

Date columns in the sales and features datasets were converted from text to datetime using the DD/MM/YYYY format. This ensured correct chronological ordering and supported calendar features and time-based evaluation.

No exact duplicate rows were found in the three source datasets. Record-key checks also found no repeated Store–Dept–Date keys in sales, Store–Date keys in features, or Store keys in stores.

Dataset Integration

Sales records were left-joined with features using Store and Date. The result was then joined with store attributes using Store. Both joins were validated as many-to-one relationships.

All sales records matched the corresponding feature and store records. The two IsHoliday columns agreed, so the repeated holiday column was removed.

The merged dataset retained all 421,570 sales observations and contained 16 columns before feature engineering. Sales totals were also checked to confirm that merging did not change the original total.

Missing-Value Assessment

The original sales and stores datasets contained no missing values. The full features dataset contained missing markdown, CPI, and unemployment values.

After restricting features to matching sales observations, CPI and unemployment were complete. Markdown missingness remained substantial:

- Markdown1: 270,889 missing values — 64.26%.
- Markdown2: 310,322 missing values — 73.61%.
- Markdown3: 284,479 missing values — 67.48%.
- Markdown4: 286,603 missing values — 67.98%.
- Markdown5: 270,138 missing values — 64.08%.

These counts refer to merged department-week records. Store-week markdown information is repeated across departments, so the counts do not represent distinct missing promotion events.

Handling Missing Information

Original missing values were preserved in the prepared dataset, and binary markdown missingness indicators were created.

For forecasting models, numerical missing values were handled within preprocessing pipelines using training-data medians and missingness indicators. Validation and test observations did not determine these replacements.

Zero and Negative Sales

The sales dataset contained 1,285 negative sales observations and 73 zero-sales observations.

Negative amounts may reflect returns or accounting adjustments, but this interpretation could not be verified from the available data. These records were retained rather than automatically deleted or changed to zero.

Weekly Continuity and Anomaly Handling

Records were sorted by Store, Dept, and Date. Gaps between consecutive observations were examined within each sales series. Missing weekly observations were not treated as zero sales.

Historical sales features were created using exact date matches. This prevented an earlier available record from being mistaken for the immediately preceding week.

Anomaly detection produced review flags rather than automatic deletions. Unusual sales may represent genuine seasonal activity, promotions, returns, or data issues. Without verified error labels, flagged observations were preserved for investigation.

Preserving the Data

Original CSV files were left unchanged. A separate compressed prepared-data checkpoint was saved with calendar features and markdown missingness indicators.

5. Exploratory Data Analysis

Exploratory analysis was used to understand sales patterns before modeling. The analysis examined overall weekly sales, store contributions, holiday differences, monthly patterns, and store types.

Each comparison was interpreted at its appropriate level. Total store sales, weekly company-wide sales, and average department-week sales represent different measures and should not be used interchangeably.

5.1 Overall Weekly Sales Trend

Weekly sales were summed across all observed stores and departments to examine the overall time pattern.

The chart shows pronounced peaks near the end of 2010 and 2011, suggesting recurring seasonal activity. These peaks motivate the use of calendar features and seasonal forecasting baselines.

A peak alone does not establish a data error. Holiday timing, promotions, and changes in observation coverage should be considered when interpreting unusually high totals.

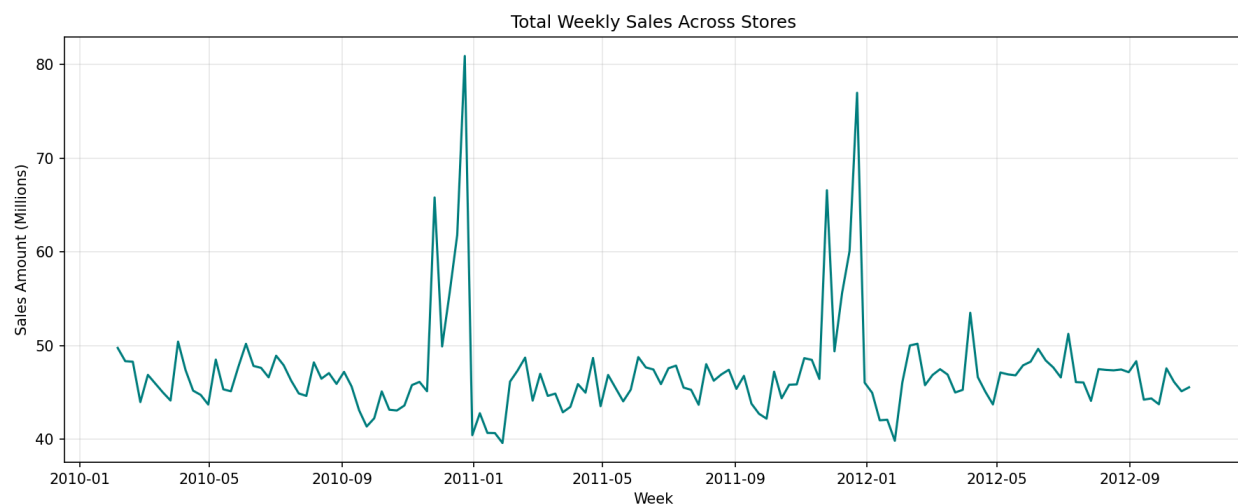


Figure 1. Total weekly sales across observed stores and departments, February 2010–October 2012. Sales amounts are displayed in millions.

5.2 Holiday Sales and Store Differences

Mean sales per observed department-week were 17,035.82 in holiday weeks and 15,901.45 in non-holiday weeks; the corresponding medians were 7,947.74 and 7,589.95. The samples contained 29,661 and 391,909 records respectively. These descriptive differences may reflect seasonality and department mix, so they do not establish a causal holiday uplift.

Stores 20, 4 and 14 were the three largest contributors by total observed sales. Total sales measure contribution, not profitability or efficiency. Store size and observation coverage must be considered before using this ranking for operational decisions.

6. Methodology and Evaluation Design

6.1 Analytical Workflow

The workflow linked validated sales, feature and store records; examined descriptive patterns; engineered calendar and historical sales inputs; detected unusual changes; grouped stores; screened department associations; and compared forecasts with simple baselines. Original

observations, negative sales and unknown markdown values were preserved. The analysis found 5,447 internal intervals longer than seven days, making exact calendar matching essential.

Sales lags were matched on Store, Dept and the exact historical date at 1, 2, 4, 8 and 52 weeks. Recent_Sales_Mean and Recent_Sales_Std summarize the available 1-, 2-, 4- and 8-week lags; they are not statistics over every week in a continuous eight-week window. Available_Recent_Lags records how much history is present. Calendar inputs include year, month, ISO week, quarter and the holiday flag.

6.2 Chronological Validation and Leakage Controls

For one-week forecasting, the first 117 observed weeks formed training, the next 13 validation, and the final 13 testing. The initial training and validation samples contained 344,667 and 38,373 observations. Model selection used validation MAE; the selected pipeline was then refitted on training plus validation before the held-out test.

Numerical imputation used training medians and missingness indicators. Store, Dept and Type were ordinal-encoded and explicitly treated as categorical inputs by the boosting model. Unseen category values were encoded as -1. Preprocessing was fitted within each training pipeline. Forecast features exclude current target sales and use only history available at the forecast origin.

The 13-week experiment used a separate origin–target design: each target is exactly 13 weeks after its origin. Training targets were restricted to be no later than the earliest validation origin, and validation targets were no later than the earliest test origin. This prevents labels from beyond the forecast origin entering training or model selection. The direct-model training and validation samples contained 229,809 and 37,954 records.

6.3 Performance Measures

$MAE = \text{mean}(|\text{actual} - \text{prediction}|)$; $RMSE = \text{square root of mean squared error}$; $WAPE = 100 \times \text{sum}(|\text{actual} - \text{prediction}|) / \text{sum}(|\text{actual}|)$. MAE and RMSE are in sales-amount units, and WAPE is a percentage. Absolute actuals in the denominator accommodate retained negative sales. WAPE can still become very large for low-volume subsets.

Both horizons use rolling forecast origins. One-week predictions use the preceding observed week as it becomes available; this is not a 13-week path issued at a single starting date. Likewise, a 13-week-ahead prediction is for one target week, not cumulative sales over the next 13 weeks. Cross-horizon metrics use different eligible observations and are not a controlled like-for-like comparison.

7. Anomaly Detection

7.1 Robust Weekly-Change Detector

For each store–department series, weekly change equals current sales minus sales exactly seven days earlier. The detector learns the median change and median absolute deviation (MAD) from training only. Its score is $|\text{change} - \text{training median change}| / (1.4826 \times \text{training MAD})$. A score above 5 triggers review. Series with fewer than 20 valid training changes or zero MAD, and records without a usable change, remain unscored.

Across the full history, 10,240 observations were flagged for review, 400,635 were within threshold, and 10,695 were not scored. Training-period flags are retrospective; later scores use frozen training statistics. Among scored records, holiday weeks had 3,553 flags out of 29,025 records (12.24%), compared with 6,687 out of 381,850 (1.75%) for non-holiday weeks. These rates support checking seasonal context before treating a flag as an error.

On the same 37,475 scorable validation records, thresholds of 4, 5 and 6 produced 692, 453 and 332 alerts respectively (1.85%, 1.21% and 0.89%). This quantifies review workload; it does not measure detection accuracy.

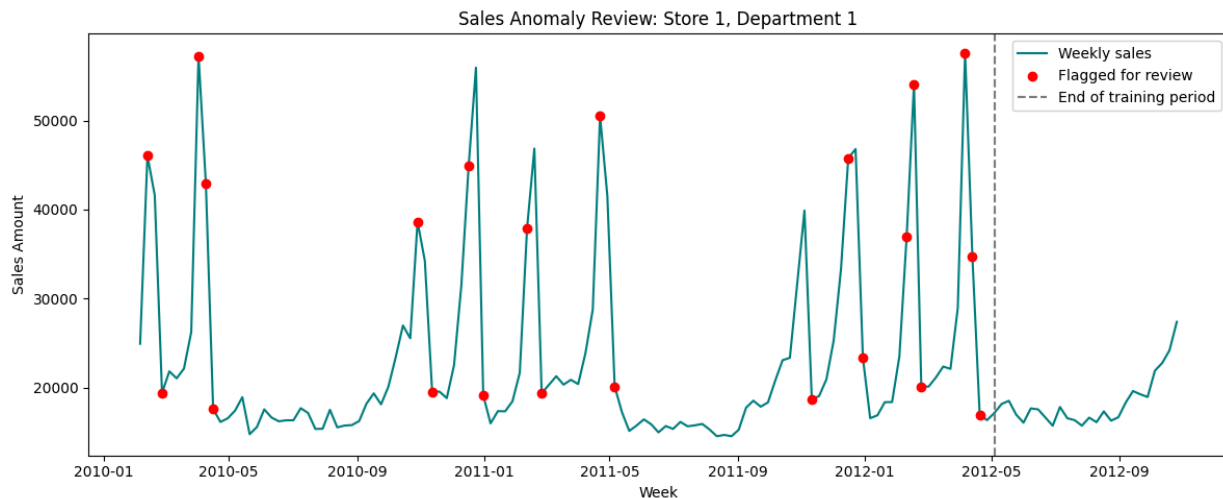


Figure 2. Original notebook anomaly plot for Store 1, Department 1. The 22 highlighted observations are review flags, not confirmed errors; the dashed line marks the training boundary.

7.2 Isolation Forest and Operational Interpretation

Isolation Forest used sales, sales change, temperature, fuel price, CPI, unemployment and store size, with training-median imputation. The model used 100 trees, contamination 0.01 and random seed 42. It flagged 358 of 38,373 validation observations (0.93%). Contamination is an assumed training threshold setting, not measured prevalence or accuracy.

Among the 37,475 records both methods could score, only 9 were flagged by both methods; 444 were flagged only by the robust detector and 349 only by Isolation Forest. Their different inputs explain why agreement is limited. Neither detector supplies ground truth. Use the

exported top-100 robust-score shortlist to investigate returns, promotions, seasonal changes and data quality while retaining the original sales values.

8. Store Segmentation

8.1 Features and Cluster Selection

Training-period sales were aggregated to store-week totals. Each of the 45 stores was represented by mean weekly sales, coefficient of variation (standard deviation divided by absolute mean sales), holiday/non-holiday mean sales ratio, and store size. StandardScaler put these four features on comparable scales. K-Means tested $k = 2$ through 6 with 20 initializations and random seed 42.

Two clusters had the highest silhouette score, 0.4567, with Davies–Bouldin index 0.8219 and a smallest cluster of 10 stores. Repeating initialization with seeds 0, 10, 21 and 100 produced adjusted Rand index 0.9011 relative to the selected solution; seed 42 reproduced it exactly. This supports reasonable initialization consistency but does not establish stability across time or samples.

8.2 Segment Profiles and Planning Implications

Segment 0, “Larger stores, higher sales,” contains 35 stores. Mean weekly store sales are 1,228,139.10; mean sales CV is 0.1545; mean holiday ratio is 1.0963; and mean recorded size is 156,223.57. It contains 20 Type A and 15 Type B stores.

Segment 1, “Smaller stores, lower variability,” contains 10 stores. Mean weekly store sales are 418,158.21; mean sales CV is 0.0922; mean holiday ratio is 1.0185; and mean recorded size is 39,511.70. It contains 2 Type A, 2 Type B and 6 Type C stores. These are descriptive store groups, not customer segments or inferred geographic regions.

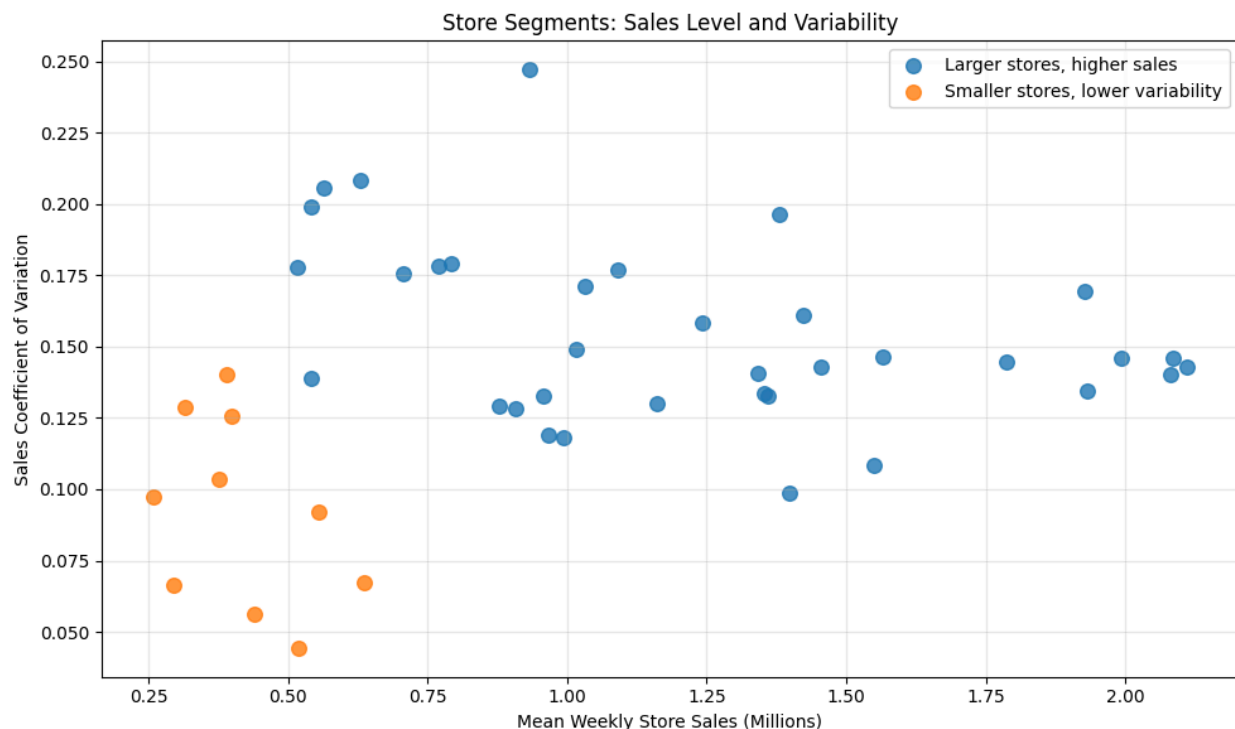


Figure 3. Original notebook store-segmentation plot. The display shows sales level and variability; clustering also used store size and holiday ratio.

On the common one-week test sample, Segment 0 had MAE 1,419.12 and pooled WAPE 8.14% across 31,385 records, while Segment 1 had MAE 600.45 and WAPE 7.29% across 6,624 records. Signed bias was +0.17% and -0.26% respectively. Low aggregate bias does not eliminate individual department errors.

9. Department Associations

9.1 Event Definition and Screening

Transaction baskets are unavailable, so this analysis examines simultaneous strong-sales events rather than joint purchases. An event occurs when department sales exceed 120% of the same store-department sales exactly 52 weeks earlier. Only non-holiday training weeks with a positive seasonal reference are eligible. The event matrix contains 2,700 store-weeks and 78 departments; unavailable observations remain missing.

Each department pair is evaluated on jointly observed store-weeks. Training screening requires at least 200 eligible store-weeks and 20 joint events. Support is the joint-event proportion, confidence is the consequent event rate conditional on an antecedent event, and lift divides confidence by the consequent's baseline event rate. The notebook generated 4,376 directed rules.

9.2 Validation Findings

The ten highest-ranked training rules were evaluated on later non-holiday validation weeks without redefining the events. Six directed rules, representing three pairs, passed the exploratory review criteria: at least 100 eligible store-weeks, 10 source events, 5 joint events and lift greater than 1.

Departments 91 and 97 had 37 joint validation events in 572 eligible store-weeks and lift 7.668. Confidence was 61.67% for 91 → 97 and 80.43% for 97 → 91. Departments 95 and 97 had 11 joint events and lift 7.199. Departments 8 and 40 had only 5 joint events; their lift fell from 20.769 in training to 7.065 in validation, illustrating the fragility of high-lift sparse rules.

The 4/95 and 2/46 pairs each had only four joint validation events and failed the minimum joint-event criterion despite lift above 1. Treat all retained rules as hypotheses for further investigation. Shared trends, store mix and promotions can create associations; no statistical significance, causality or cross-selling benefit was established.

10. Forecasting Results

10.1 One-Week-Ahead Model

The selected core model was HistGradientBoostingRegressor with 100 iterations, learning rate 0.08, maximum 31 leaf nodes, L2 regularization 10, early stopping disabled and random seed 42. Its 17 inputs combine store attributes, target calendar information and historical sales. The last-week baseline repeats sales seven days earlier; the seasonal baseline uses the 52-week lag and falls back to last-week sales when needed.

On the common 38,009-observation test sample, the selected core model achieved MAE 1,276.44, RMSE 2,652.42 and WAPE 8.06%. The last-week baseline achieved 1,525.50, 3,444.10 and 9.63%; the seasonal baseline achieved 1,693.57, 3,628.49 and 10.70%. Core-model MAE was approximately 16.3% lower than the last-week baseline.

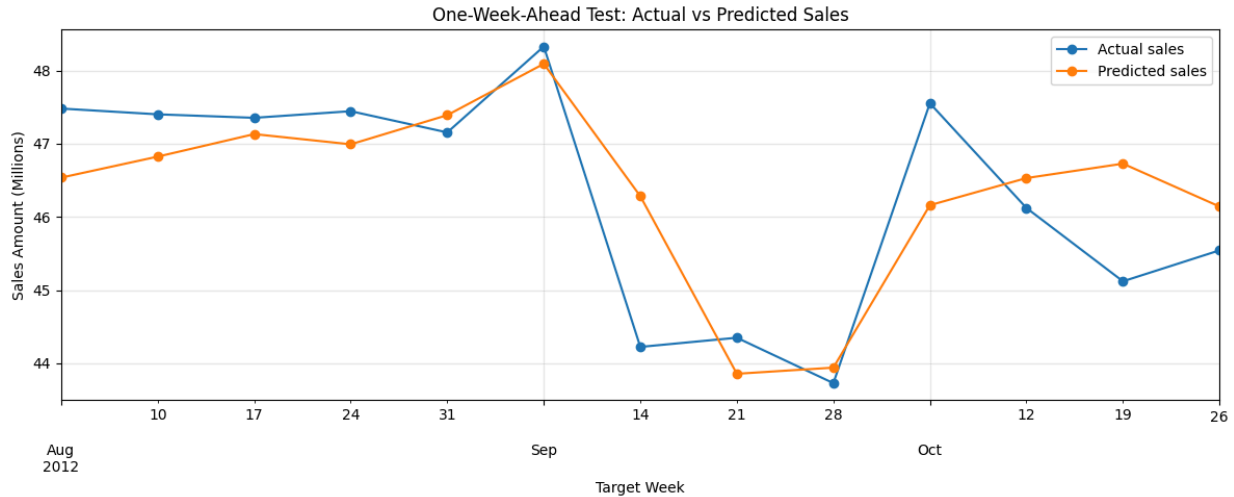


Figure 4. Original notebook one-week test chart: weekly totals on the common comparison sample. Aggregation can cancel errors, so department-level metrics remain necessary.

Coverage matters: the core model scored all 38,530 test records, achieving MAE 1,268.77, RMSE 2,640.73 and WAPE 8.12%. The 521 records missing previous-week sales were excluded from the common baseline comparison; separately they had MAE 708.89 and WAPE 625.71%. This low-volume, missing-history subset needs exception handling rather than reliance on the headline score.

10.2 Thirteen-Week-Ahead Model Selection

On 37,954 validation records, seasonal-baseline MAE was 2,017.54, compared with 2,088.43 for direct 13-week boosting and 4,602.09 for the origin-sales baseline. The boosting model had lower validation RMSE (4,266.38 versus 4,498.90), but the predefined selection criterion was MAE, so the seasonal baseline was retained.

On 37,641 test observations, the selected seasonal baseline achieved MAE 1,701.58, RMSE 3,640.31 and WAPE 10.67%. The origin-sales baseline achieved MAE 2,788.14, RMSE 6,135.64 and WAPE 17.48%. For 587 records without the target's 52-week seasonal reference, the selected method fell back to observed origin sales.

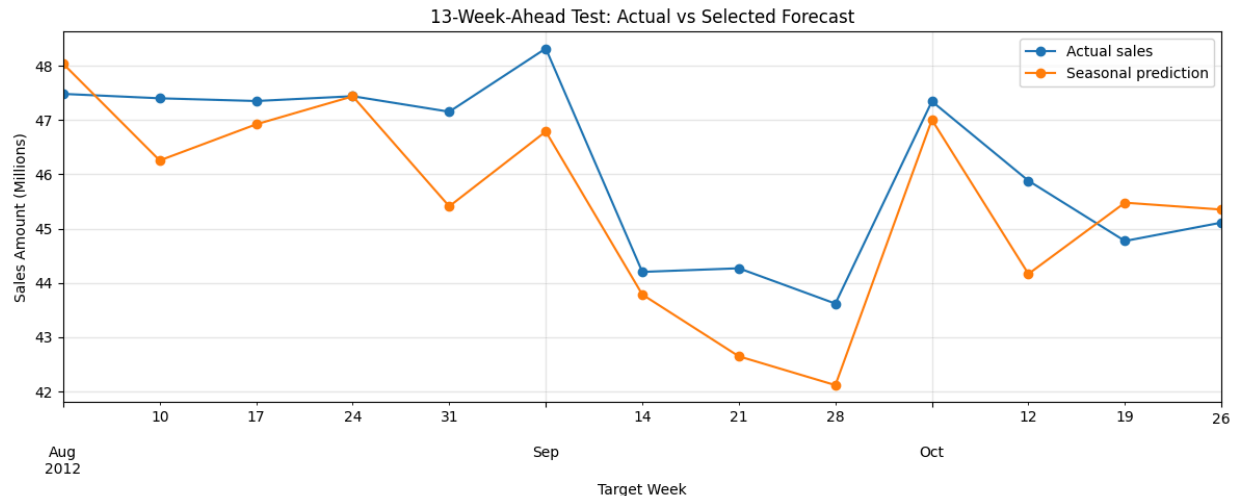


Figure 5. Original notebook 13-week-ahead test chart: actual and selected seasonal forecasts aggregated by target week. Each prediction uses its own rolling origin; this is not a cumulative 13-week demand forecast.

10.3 Saved Future Forecasts

The future one-week pipeline was refitted on all available sales history after evaluation. Both future exports cover the 2,959 store–department pairs observed in the latest sales week, rather than every historically observed pair. These are forecasts relative to the historical dataset cutoff, not current retail forecasts.

One-week target: Nov 2, 2012. Predicted sales sum to 43,703,343.31 for the larger-store segment and 4,312,620.51 for the smaller-store segment.

Thirteen-week target: Jan 25, 2013. Predicted sales sum to 35,868,089.49 and 3,978,838.08 respectively. These totals concern different target weeks and methods; their difference is not a growth estimate. Values are sales amounts, not product quantities.

11. External Factors

The extended forecasting experiment added nine one-week-lagged variables: temperature, fuel price, CPI, unemployment and Markdown1–Markdown5. Using historical values avoids assuming that realized target-week external conditions were known in advance.

On the same validation comparison sample, core gradient boosting achieved MAE 1,354.35, RMSE 2,731.07 and WAPE 8.29%. Adding lagged external features produced MAE 1,394.00, RMSE 2,785.39 and WAPE 8.54%, so the extension was not selected. This result concerns the tested features, lag and model settings; it does not show that external factors have no economic effect.

For descriptive segment context, feature records were deduplicated to one observation per store-week and then averaged with equal weight per store. Markdown availability averaged 20.82% for larger stores and 16.22% for smaller stores in training. Means of recorded

markdown values exclude unknown weeks and must not be interpreted as average spending across all weeks. Differences in temperature, CPI and unemployment are contextual, not evidence of geography or causal sales drivers.

12. Business Recommendations

Larger stores: review forecasts and large sales changes weekly, with attention to holiday periods and greater observed variability. Check inventory and supplier lead times before changing orders. Use the one-week model as a planning input alongside exception review; retain seasonal baselines for comparison and longer-horizon planning.

Smaller stores: use a regular replenishment review cycle with alerts for deviations. Their lower average variability supports a simpler review cadence, but missing-history departments still need manual attention. Do not derive reorder quantities or safety stock directly from monetary sales forecasts.

Marketing: test holiday-timed campaigns in larger stores and smaller targeted campaigns in the lower-variability segment. Use comparable holdout stores and evaluate incremental margin after promotion costs. The observed holiday ratios and department rules motivate pilots; they do not quantify a proven campaign benefit.

Anomaly review: prioritize unusually large changes and records flagged by both detectors, then investigate the surrounding weeks, returns and promotion records. Record reviewer outcomes so precision and recall can eventually be measured. Do not delete records solely because an algorithm flagged them.

Department associations: first examine the 91/97 and 95/97 pairs using transaction-level evidence and department definitions. Avoid product-bundle or cross-selling claims from weekly co-movement alone. Monitor forecasting MAE, WAPE, coverage and signed bias by store and segment before expanding operational use.

13. Limitations and Further Work

Data limitations: the records contain sales amounts, not unit demand, customer identities, transaction baskets, inventory, margins, promotion costs or supplier lead times. Missing markdowns remain unknown; store-size units and detailed department meanings require source documentation. Negative sales and gaps were retained because their underlying causes were not verified.

Validation limitations: reported accuracy comes from one chronological split per horizon and rolling-origin evaluations, without prediction intervals or broader repeated historical backtesting. The final test target period does not include a year-end holiday peak. Future performance, peak-season reliability and cold-start behavior therefore require additional validation.

Analytical limitations: anomaly labels are unavailable; alert thresholds are exploratory. K-Means used only 45 stores and was checked for initialization sensitivity, not future stability. Association

screening evaluated many pairs without significance testing and may reflect common trends. External-feature comparisons are predictive and do not estimate causal effects.

Next steps are to collect unit sales, inventory, lead times, margins, promotion costs and transaction records; verify business definitions; add rolling historical backtests and uncertainty intervals; review sparse or missing-history series; and evaluate inventory and marketing decisions through controlled pilots. These additions are proposed work, not results of the current notebook.

The verified outcome is a reproducible historical analysis with differentiated store profiles, review-oriented anomaly detection and horizon-specific forecasting choices. The one-week core model improves on the tested baselines, while the 13-week seasonal baseline remains the MAE-selected approach.